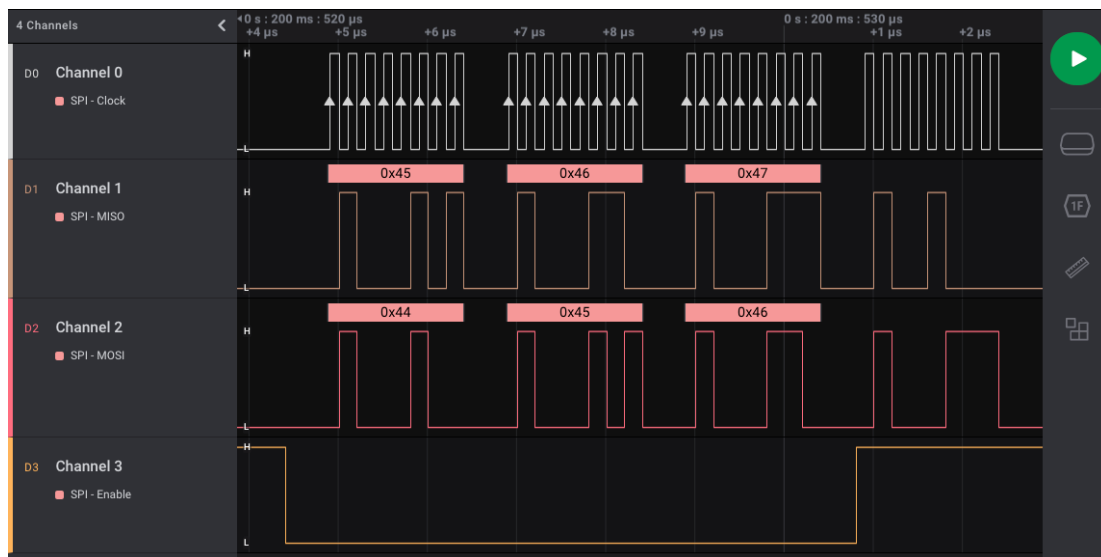


SPI Analyzer - User Guide

The Saleae software includes a protocol analyzer for the Serial Peripheral Interface (SPI) bus.



SPI Decoding in the Logic 2 Software

SPI uses a Clock signal, two data signals (MISO and MOSI), and an Enable signal. That is the most common configuration of SPI, but other variants exist.

Our SPI analyzer is generic enough to decode basic synchronous serial (at minimum, a clock signal and a data signal) without the need for a second data signal or an enable signal. In cases like this, one of the data signals and the Enable signal can be disabled from within the analyzer settings.

Setup

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Add the SPI analyzer to your capture. There are a number of settings for the analyzer. First, select the channels you are using. If you are not using an enable signal or if you only have one data signal, simply set the unused channels to "None". Only clock and one data channel (either MISO or MOSI) are required for the SPI analyzer to operate.

Next, select the remaining settings you are using. If you are unsure what settings are used for the data you captured, you can either visually identify them or you can guess and check.

- MSB/LSB first. If you can't check which mode your data uses, you may need to guess.
- Bits per transfer. This is almost always 8 bits but can be set up to 64 bits.
- Clock polarity

Clock polarity is basically the idle state of the clock in between SPI transactions. Whatever the idle clock state is when the enable signal becomes active, use this value (high or low).

- Clock phase

Clock phase selects which edge the data is read on. Please note that leading and trailing do not correspond to rising and falling, which can lead to confusion. The leading edge is defined as the first edge on the clock signal after the enable line becomes active. If the clock is idle high, then the leading edge is the falling edge. If the clock is idle low, then the leading edge is the rising edge. This is the edge where data will be read from the data lines.

- Active low/active high enable

Only applies when an enable signal is used. Defines the state of the enable signal when the SPI analyzer will read the data signals.

Multiple Devices / Enable Signals

When you have multiple enable signals, one for each device, simply add more SPI analyzers. If you have the same clock, the MISO and MOSI channels are already

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Different Settings for MISO and MOSI

In some cases, you may want to have different settings between the two data channels. That can easily be done by using two SPI analyzers. Analyzers in the Saleae software have no limit to the number of analyzers that can share the same channels, so just add two SPI analyzers using the same clock, and enable channels. Then, on each analyzer, disable one of the two data channels.

SPI analyzer 1

- MOSI channel - MOSI input selected
- MISO channel - "none"

SPI analyzer 2

- MOSI channel - "none"
- MISO channel - MISO input selected

Then assign the settings you like to each analyzer.

Using the SPI Analyzer Without an Enable (Chip Select) Signal

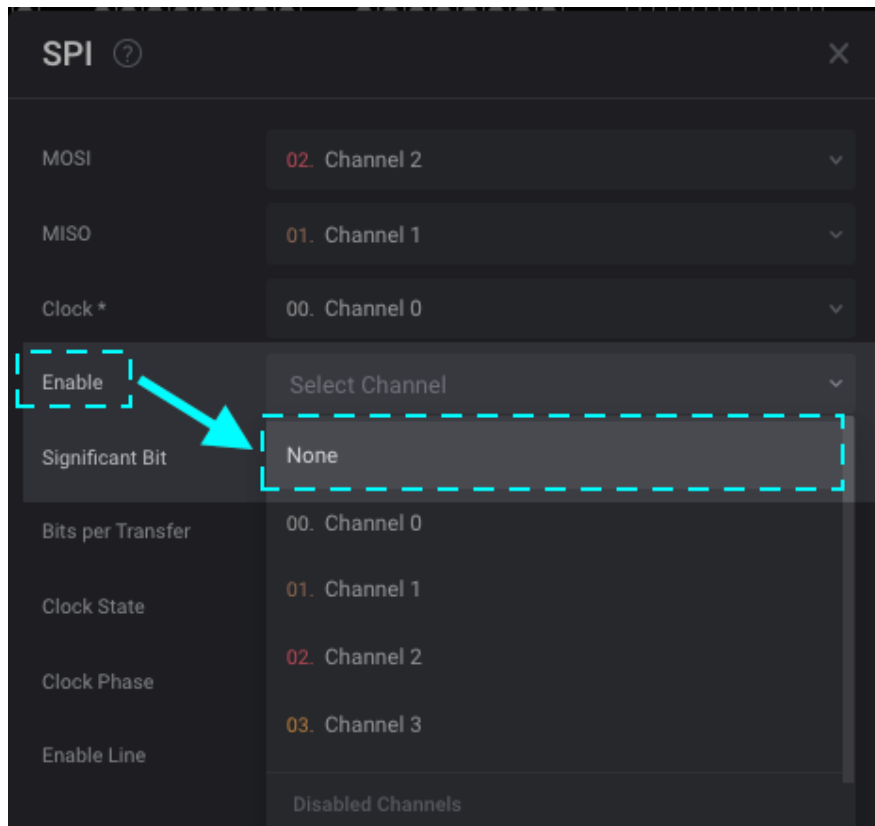
In many cases, there is no chip select signal available for an SPI bus that needs to be recorded.

The SPI analyzer included in the Saleae Logic software supports this case. In the event you record SPI data and there is no valid enable signal, simply change the "Enable" channel in the SPI analyzer settings to "None" as shown here:

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Setting Enable to "None"

Solving Alignment Issues

In some cases (most commonly caused by ignoring the enable signal as described above), the analyzer will not decode the data correctly. That is because the SPI analyzer usually relies on the enable signal to align the data—that is, determine which bit of data should be the first bit in each byte and which bit should be the last.

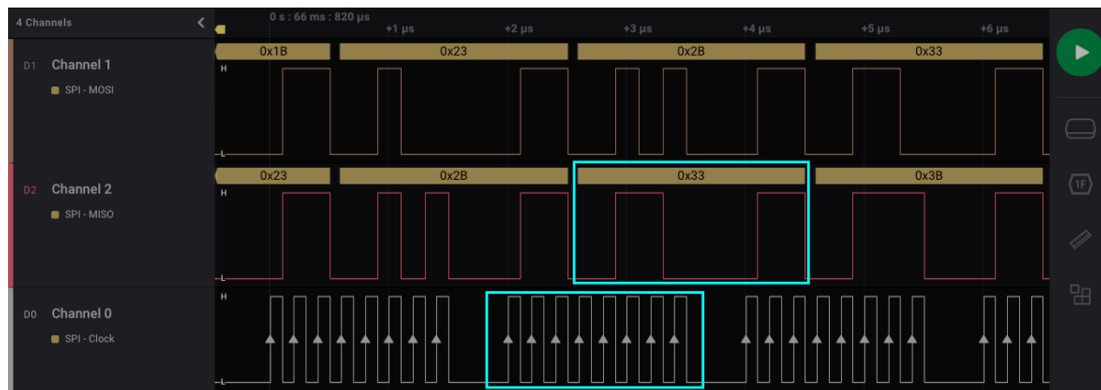
Shown here is a typical issue where the alignment is off. You can clearly see that the pulses on the clock channel are in groups of eight—8 bits in each byte. There is a short gap between each byte.

However, even though starting on the correct previous byte and address

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SPI Analyzer Alignment Issue

Here are the two most common causes of this:

- The logic analyzer started recording in the middle of an SPI byte.
- There are other erroneous transitions on the clock channel before the valid SPI data.

This is easy to fix. You will need to tell the SPI analyzer where the valid SPI data starts in the capture. The analyzer will then use that as the new starting point of the capture and decode data from that location.

Here's how this is done:

1. Place a timing marker just before the start of the first valid SPI byte.
2. Delete the data before the timing marker (**please note this cannot be undone**). Instructions are provided in the support article below.

Delete Part of your Capture

Logic 1.x

If using the older Logic 1.x

Solving Alignment

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